

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and optimized **ETL/ELT pipelines** using **Python** and **SQL** (PostgreSQL, MySQL, Oracle) to process enterprise financial data; built **data quality controls** and validation workflows reducing data errors by **40%** and enabling reliable downstream analytics across 50+ banking operations dashboards.
- Designed and deployed **Databricks Lakehouse** solutions leveraging **Delta Live Tables** for real-time data ingestion and transformations on banking transaction flows; implemented **Unity Catalog** IAM policies and access control frameworks ensuring **100% GDPR and regulatory compliance** across financial services datasets.
- Built **AWS cloud infrastructure** for data platform using **S3, Lambda, and CloudWatch**; managed **Terraform**-based resource provisioning and infrastructure-as-code pipelines; automated monitoring and alerting for data pipeline performance, reducing incident detection time by **65%** and maintaining **99.8% platform SLA**.
- Implemented **CI/CD pipelines** with **Git** version control and **Jenkins** automation for data engineering workflows; conducted code reviews and integrated **data quality monitoring solutions** into deployment processes, achieving **zero critical production data issues** across 12 concurrent pipeline deployments.
- Collaborated with **data scientists, analysts, and business stakeholders** in **Agile/Scrum ceremonies** to translate raw financial data into actionable insights; documented data governance standards, API integration specifications, and **technical documentation** for enterprise data platform supporting multi-currency payments and cross-border settlement operations.
- Applied **Spark programming** and **Python data stack** (pandas, NumPy, scikit-learn) to develop scalable data transformations processing **500GB+ daily transaction volumes**; leveraged **AI tools** to optimize query performance and accelerate development velocity by **30%**, demonstrating continuous learning of emerging data engineering technologies.

SKILLS

Data Engineering & Lakehouse – Databricks, Delta Lake, Delta Live Tables, Unity Catalog, Lakehouse architecture, ETL/ELT pipeline development, data modeling, data quality controls, data governance.

Cloud & Infrastructure – AWS (S3, Lambda, CloudWatch, Glue), Terraform, infrastructure-as-code, resource management, CI/CD (Jenkins, Git), monitoring solutions.

Programming & Data Processing – Python, Spark (PySpark), SQL (PostgreSQL, MySQL, Oracle), pandas, NumPy, scikit-learn, API integrations, real-time processing.

Financial Services & Compliance – Banking domain expertise, SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA, cross-border payments, PSD2, GDPR, regulatory compliance, financial data security.

Collaboration & Development – Agile/Scrum methodologies, version control (Git), technical documentation, stakeholder communication, Jira, change-control management.

Data Analytics & Visualization – Power BI, Tableau, DAX, data warehousing, exploratory data analysis, statistical methods, machine learning fundamentals.

EDUCATION AND TRAINING

PROJECTS

Enterprise Centralized Data Platform – Databricks Lakehouse Migration (HCL Technologies)

- Led design and deployment of **Databricks Lakehouse Platform** consolidating legacy data silos from banking operations, payments systems, and ERP modules; architected **Delta Live Tables** ingestion framework ingesting **2TB+ daily transaction data** from 15+ source systems using **Python** and **Spark programming**.
- Implemented **Unity Catalog** governance framework with role-based access control and data lineage tracking; applied **IAM best practices** and **AWS S3, Lambda, Glue** integration patterns to enforce GDPR and financial regulatory standards across **100+ analytics datasets**, achieving **zero compliance violations** during audit.
- Developed **data quality monitoring** and alerting pipelines using **CloudWatch** and custom **Python validation modules**; automated **Terraform**-based infrastructure provisioning reducing platform deployment time by **50%** and enabling self-service analytics consumption for 40+ business stakeholders.
- Collaborated across data engineering, data science, and business operations teams in **Agile/Scrum sprints**; delivered comprehensive **technical documentation**, API integration guides, and training materials for **Delta Lake, Unity Catalog, and real-time processing** capabilities supporting ISO 20022 message flows and cross-border payment reconciliation.
- Optimized **ETL/ELT pipeline performance** through **Spark query optimization** and **SQL** tuning, reducing end-to-end data latency by **70%**; leveraged **AI-assisted development tools** for rapid prototyping and root-cause analysis, accelerating feature delivery velocity by **35%** across 6-month implementation.

SWIFT Payments Operations Dashboard & Data Pipeline Modernization (HCL Technologies)

- Built and maintained production **data ingestion pipeline** for SWIFT messaging dashboard processing **100K+ daily payment transactions**; implemented **SQL**-based transformations (PostgreSQL, Oracle) and **Python ETL scripts** to normalize **MT/MX message formats** and ISO 20022 structures for operational visibility and transaction-flow traceability.
- Designed **data quality framework** with automated reconciliation validation and anomaly detection; partnered with banking operations and regulatory teams to ensure **GDPR compliance**, audit trail completeness, and SLA breach prevention, reducing payment discrepancies by **45%** and enabling real-time operational alerting.
- Supported **Git-based CI/CD workflows** and **change-control processes** for pipeline deployments; documented technical specifications and integration patterns for cross-border payments, SEPA settlement, and PSD2-compliant data flows, enabling seamless UAT and post-implementation support across interconnected financial systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)